

Drift-Sense

Citations & Literature Justification

SEMICON India 2026 Hackathon | Applied Materials Problem Statement PS-2

Team: The Nanoscale Collective

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Why This Document Exists

The problem statement requires that structures, noise, and augmentations used in the synthetic dataset be justified with at least 2-3 credible public sources. This document goes a step further: rather than a flat reference list, each source below is tied directly to the specific parameter or design decision in `dataset_generator.py` it justifies, so the connection between literature and code is explicit and checkable.

All sources are real, peer-reviewed or standards-body publications, verified individually rather than assumed. No confidential fab data or proprietary process information was used anywhere in this project — every structural and noise assumption traces back to a source below.

1. Sensor Noise — Poisson-Gaussian Signal-Dependent Model

Foi, A., Trimeche, M., Katkovnik, V., & Egiazarian, K. (2008). Practical Poissonian-Gaussian noise modeling and fitting for single-image raw-data. *IEEE Transactions on Image Processing*, 17(10), 1737–1754.

DOI: <https://doi.org/10.1109/TIP.2008.2001399>

This paper models imaging-sensor noise as a signal-dependent combination of a Poissonian component (photon/electron counting statistics) and a stationary Gaussian component, giving pointwise noise standard deviation as a function of the underlying signal: $\sigma(x) = \sqrt{a \cdot x + b}$. This is a physically grounded alternative to flat additive Gaussian noise, and matches how electron-counting noise actually behaves in SEM-style imaging.

Used for: *The Poisson shot noise component in `dataset_generator.py` 's degradation pipeline, implementing $\sigma(x) = \sqrt{a \cdot x + b}$ directly rather than assuming uniform noise across all intensity levels.*

2. SEM Charging Artifacts

Cazaux, J. (1986). Some considerations on the electric field induced in insulators by electron bombardment. *Journal of Applied Physics*, 59(5), 1418–1430.

DOI: <https://doi.org/10.1063/1.336493>

A foundational physical treatment of why insulating regions on a sample accumulate charge under electron-beam bombardment, and how the resulting local electric field distorts the resulting image — the physical basis for the streak-like brightness artifacts commonly seen in real SEM images of semiconductor structures with exposed dielectric regions.

Used for: *The charging-streak degradation in `dataset_generator.py` — justifies modelling this as a distinct artifact from generic sensor noise, since its physical cause (local field distortion from trapped charge) is different from photon/electron counting noise.*

3. DRAM / Fin-FET Pitch and Structure Scaling

IEEE International Roadmap for Devices and Systems (IRDS), 2023 edition, Metrology chapter, Table MET-2 (“Logic Device Structure Requirements,” technology years 2022–2037).

Direct PDF: https://irds.ieee.org/images/files/pdf/2023/2023IRDS_MET.pdf

The IRDS (successor to the ITRS) is the semiconductor industry's consensus public roadmap for logic and memory technology scaling. Table MET-2 specifically tabulates projected logic device structure requirements — including Fin-FET and gate-all-around geometry — across technology years 2022 through 2037, giving a checkable, year-by-year anchor rather than a general appeal to the roadmap's authority. It is explicitly public, non-confidential information — exactly the kind of source the problem statement permits and IRDS is designed to provide.

Used for: *Realistic pitch, line-width, and structural proportions in `dataset_generator.py` 's DRAM and Fin-FET pattern generators, so the synthetic layouts reflect plausible real-world scaling relationships rather than arbitrary geometry.*

4. Wafer Stage Drift and Overlay Error

Brunner, T., Menon, V. C., Wong, C. W., et al. (2014). Characterization and mitigation of overlay error on silicon wafers with nonuniform stress. Proc. SPIE 9052, Optical Microlithography XXVII, 90520U.

DOI: <https://doi.org/10.1117/12.2045715>

Shows that non-uniform residual stress across a wafer — measured on purpose-built Engineered Stress Monitor wafers via a Patterned Wafer Geometry tool — produces overlay (layer-to-layer positional) error that is structured and location-dependent, not simple random noise, and in the non-uniform case is often non-correctable by standard uniform-offset overlay correction. This establishes that navigation/positional error in wafer processing has a genuine systematic component tied to physical wafer state, not only random mechanical vibration.

Used for: *The drift/jitter component of dataset_generator.py 's degradation set — specifically justifying why drift is modelled as having a structured, non-purely-random component rather than flat Gaussian jitter alone.*

Coverage Against the Problem Statement's Suggested Degradations

The problem statement lists blur/astigmatism, shot noise, drift/jitter, contrast/gamma change, charging streaks, salt-and-pepper noise, spot-size blur, and feature loss during down sampling as possible degradations, explicitly noting no single noise model is mandatory as long as choices are technically justified. The table below maps our implemented degradations to their literature grounding.

Degradation	Implemented?	Grounded By
Shot noise	Yes — Poisson-Gaussian	Foi et al. 2008
Charging streaks	Yes	Cazaux 1986
Salt-and-pepper noise	Yes	Standard sensor defect model
Contrast / gamma change	Yes	Standard SEM detector response variation
Drift / jitter	Yes	Wafer overlay error literature
Blur / astigmatism	Yes — blur; astigmatism-specific term not separately modelled	Noted as a scope decision, see Known Limitations in README
Spot-size blur	Partially covered under general blur	Noted as a scope decision
Feature loss during down sampling	Implicit in the 10x scale reduction from reference to search	Standard down sampling behaviour

Honest note: astigmatism and spot-size blur are not modelled as physically distinct effects from general Gaussian blur in the current generator. This is a deliberate scope decision given the project timeline, not an oversight — flagged here explicitly rather than implied to be covered.